

Name 1:

ID# 1:

Name 2:

ID# 2:

Problem 1. A student takes a 10 question true-false quiz, but did not study and randomly guesses each answer. Let X denote the number of correct answers.

- (a) which type of distribution X has? write its parameters.
- (b) Find the probability that the student passes the quiz with a grade of at least 80% of the questions correct.

Lab 4 (Group Discussion)

Name 1:

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ID# 2:

Problem 2. Suppose that a lecture note with 20 pages contains on average 4 misprints per page. What is the probability that there will be at least 8 pages which contain more than 5 missprints?

Name 1:

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Problem 3. Assume we toss a fair coin until we get either three heads or three tails, whichever comes first, then we stop. Let the random variable X be the number of times we toss the coin.

- (a) What type of probability distribution does X have?
- (b) What is the probability that $X = 5$?